

Internet Explorer

Draw and explain the client-server architecture and trace what happens when you open a website.

Client–Server Architecture: The client and server communicate over the internet using protocols like HTTP/HTTPS.

Client–Server Architecture is a network model where:

- Client → Requests services or resources
- Server → Provides services or resources



Figure 1: Diagram of Client–Server Architecture

Components:

1. Client

- The user's device (Laptop, Mobile).
- Uses browsers like Google Chrome, Mozilla Firefox, or Internet Explorer.
- Sends request to server.

2. Server

- A powerful computer that stores website files.
- Processes requests and sends responses.

3. Database

- Stores website data (user accounts, passwords, products, etc.).

4. Internet

- Medium through which communication happens.

What Happens When You Open a Website:

Step 1: URL Entered

You type the website URL in the browser.

Step 2: DNS Lookup

- Browser contacts **DNS (Domain Name System)**.
- DNS converts domain name into IP address.

Step 3: HTTP Request Sent

- Browser sends an HTTP/HTTPS request to the server.

Step 4: Server Processing

- Server receives request.
- Server processes it.
- If needed, server fetches data from database.

Step 5: HTTP Response

- Server sends back:
 - HTML
 - CSS
 - JavaScript
 - Images

Step 6: Browser Rendering

- Browser renders the page.
- Website appears on your screen.

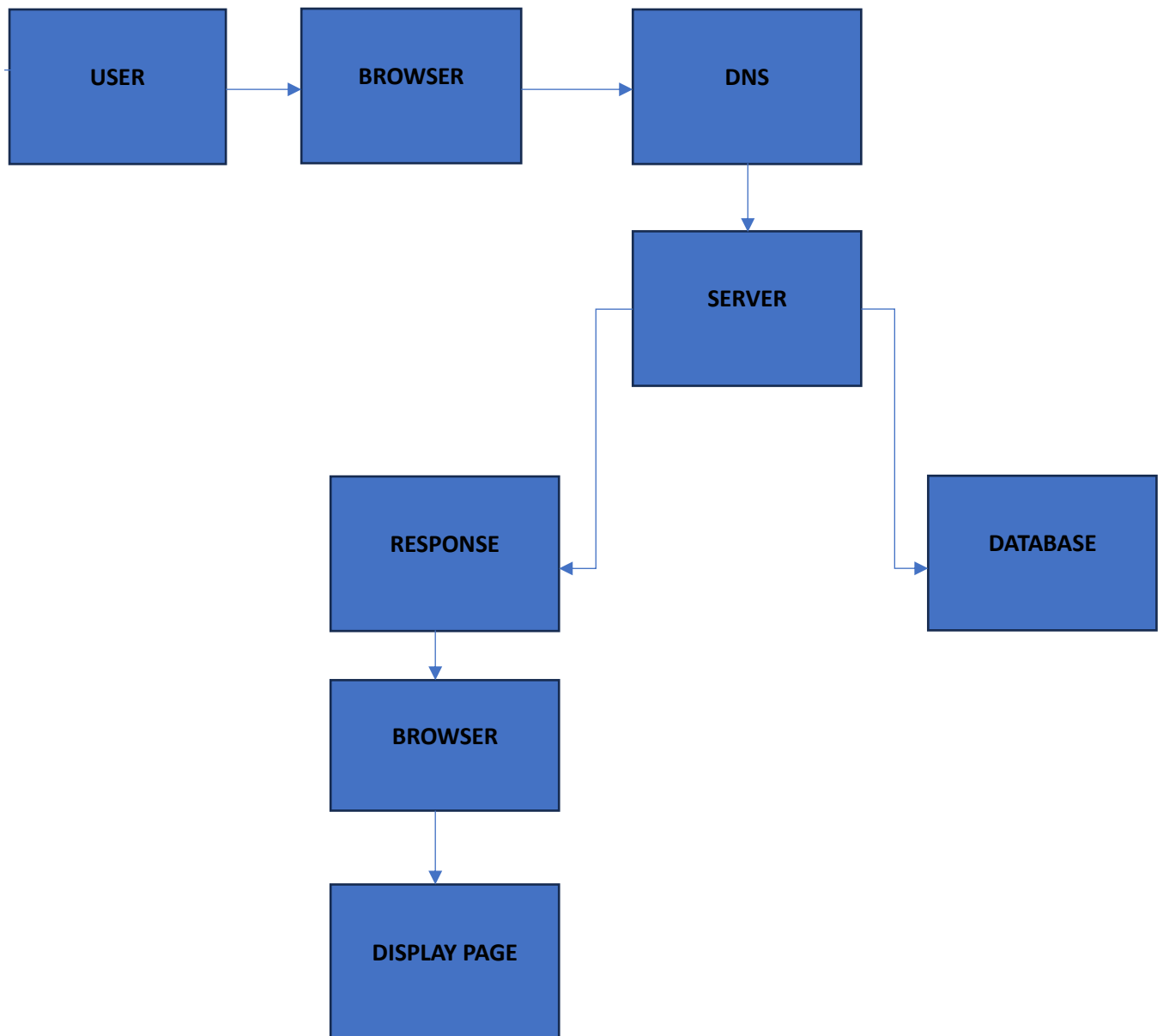


Figure 2: Simple Flow Diagram of Website Opening

Conclusion:

Client-server architecture is the basic foundation of how the Internet works. When a user opens a website, the client (browser) sends a request to the server through the Internet. The server processes the request, retrieves necessary data from the database if required, and sends back a response. The browser then displays the webpage to the user. This entire process happens very quickly and allows us to access websites easily and efficiently.